

Dam Monitoring System

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Abstract—This project report focuses on the build-up of the Dam Monitoring System and its application. Dams play an essential role in producing currents using water flow; they are also used to save people's lives on the river flow side by stopping water overflow. This system contains LED lights, which will respond according to different water levels, and a float sensor, which responds when the water from the dam touches the safety level. This report serves as a generic guide to the dam monitoring system development and application and gives valuable information to engineers, researchers for further development and research.

Index Terms—Water level monitoring, automated system, led indicator, alert system.

I. INTRODUCTION

Bangladesh is one of the riverine countries in the world. Many rivers, lakes, and ponds surround it, which has made Bangladesh a disaster-prone country, too. Bangladesh shares 57 international rivers with its neighbors. Especially in the monsoon season, riverside areas face severe floods. Because of the flood, Sylhet, Sunamganj, Kurigram, Jamalpur, and Rangpur are the most effected areas.

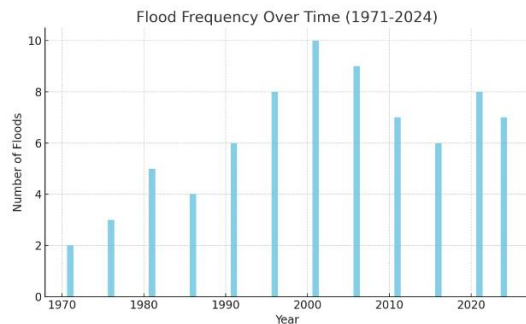


Fig. 1. Graph of flood from (1971-2024).

The first graph shows the frequency of floods from 1971 to 2024, highlighting a fluctuation with peaks around 2000 and 2020.

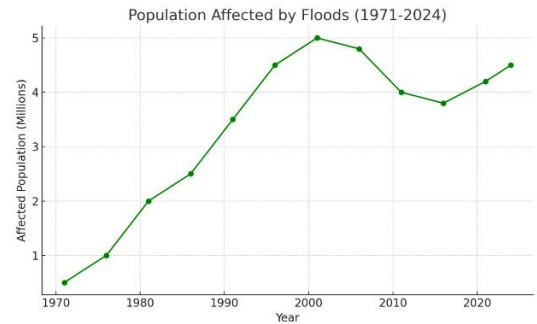


Fig. 2. Graph of population affected by flood .

The second graph shows the population affected by floods during the same period, indicating a significant rise until 2000, after that a slight decline and a recent increase.

II. LITERATURE REVIEW

One of the papers in this field was "Dam Management and Disaster Monitoring System using IoT"[1]. The authors proposed IoT-based dam monitoring system that uses various sensors to collect data on rain, water level, and temperature. The system detects real-time monitoring and give notifications via alarm triggers.

In "Design of Flooding Detection System Based on Velocity and Water Level DAM with ESP8266"[2], the authors proposed a wireless sensor-based dam monitoring system using ESP8266. The system uses an application called SWOD (Save Water Overflow Detection) to get the early warning. For monitoring the water level, they use Figure App so that users can monitor the water level at anytime.

Several other papers have proposed similar projects on the dam monitoring system, such as "Real Time Monitoring and Controlling of Water Level in Dams using IoT"[3]. The authors proposed a system where all network devices are connected through IoT (Internet of Things). This system uses various sensors to collect data and store it in the Thingspeak server. Servo motors are connected to the system to open or close the dam gate.

Another article on the Dam Monitoring System is "The Development of Water Quality Monitoring System"[4]. This article focuses on a system where various sensors are used with ESP8266 NodeMCU to maintain water quality. By connecting the system with NodeMCU, the authors make a webpage to show data. They also use the Blynk app so that users can monitor the water level and be alerted at a critical time.

In "Smarter dam based on cyber-physical system utilizing Raspberry Pi4 and NodeMCU ESP8266"[5], the design is based on the connection of NodeMCU with Google Cloud. The authors stored the data in Google Cloud which they were getting with the help of NodeMCU. They also proposed using Drone to surveil the dam for its safety.

III. METHOD

The flowchart of the Dam Monitoring System is shown in figure 1. After activating the system, the water sensor will detect whether the water level is low, average, or high. If the water level is low, the green LED will flash; otherwise, the water sensor will detect the water level. If the water level is average, the yellow LED will flash, or the water sensor will detect the water level again. If the water level is high, the red LED will flash, sending a message through sim9001. The message will carry a high alert, and it also means that the dam can be opened at anytime. A buzzer will sound at an alarming level to inform the dam's surrounding area.

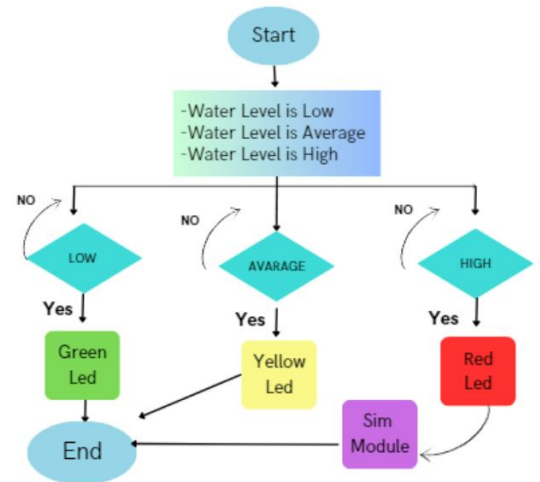


Fig. 3. Flow Chart of dam monitoring System.

A. Circuit construction

In this project we use Arduino UNO is a micro controller (brain) that controls the complete circuit. Also we use a Hc-

SR04 ultrasonic sensor, a water module , a Sim900A, a OLED display, led(red, blue green) etc. Ultrasonic sensor basically measure the distance of the water from the top of the dam, water module measure the water level, according to water level the led will glows (Green for low water level, yellow for medium water level red for high water level) and a sim module which can sent SMS when high level water were detected.

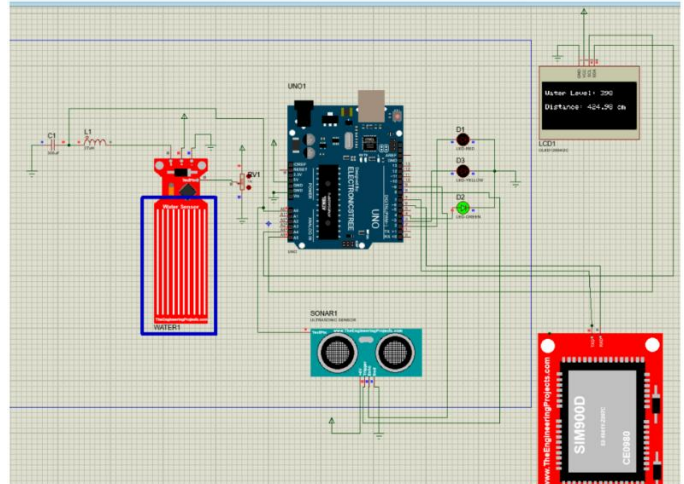


Fig. 4. Simulation of dam monitoring System.

In this simulation we use a variable resistor with the output pin of the water module sensor which refers if we increase the resistance and if we decrease the resistance then the water level is decreasing. Here water module sensor is an analogue sensor which has unstable output voltage and high frequency. So we use a capacitor and a inductor with the output pin of the water module sensor.

B. Hardware implementation

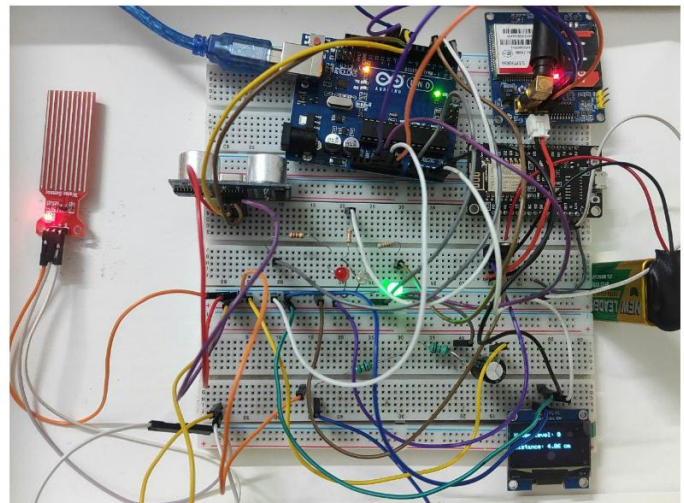


Fig. 5. Hardware of dam monitoring System.

Figure 5 shows the arduino circuit that have been constructed on the breadboard. Arduino have basically two types of

pin digital and analogue. we connect the analogue and digital devices with the analogue and digital pins of the arduino. As a power supply we also use a 9V battery. Finally we can see that the system can easily detect the water levels. And according to the water level it will turn on the leds. Also it can alert people through sms.

IV. RESULT AND DISCUSSION

Being an irrigation system, the water levels are determined by the ultrasonic and float sensors with details relayed on a OLED screen. Also for led glow we use a condition like if (water level less than 450) then green led will turn on, if (water level is between 450 and 550) blue led will turn on and if (water level greater than 550) then the red led will glow.

We faces some problems with sim800l module which has network issues. So then we use sim900a module which is perfectly work. Also we face some graphics issue in oled display and by adding library's we solve this problem.

V. CONCLUSION

The dam monitoring system is designed to solve the flood occuring problems. Thus, the project will help the residents to be ready, especially for floods. The water sensor has been implemented to monitor the dam automatically, which will also help send messages to people's mobiles. Though it is an automatic dam monitoring system, there is also human control. The ultrasonic sensor has been used carefully in this project to measure the water distance easily from time to time. With the help of a buzzer, the residents will be able to be alerted when the water level crosses the danger level.

REFERENCES

- [1] Dam Management and Disaster Monitoring System using IoT. (2023, March 23). IEEE Conference Publication — IEEE Xplore. <https://ieeexplore.ieee.org/document/10105132>
- [2] Design of flooding detection system based on velocity and water level DAM with ESP8266. (2017, November 1). IEEE Conference Publication — IEEE Xplore. <https://ieeexplore.ieee.org/document/8285537>
- [3] Real Time Monitoring and Controlling of Water Level in Dams using IoT. (2018, December 1). IEEE Conference Publication — IEEE Xplore. <https://ieeexplore.ieee.org/document/8692099>
- [4] <https://bmi.unikl.edu.my/wp-content/uploads/2021/11/82-85-The-Development-of-Water-Quality-Monitoring-System-at-Dam20210617b.pdf>.
- [5] Deab, M. Y., Croock, M. S. (2022). Smarter dam based on cyber-physical system utilizing Raspberry Pi4 and NodeMCU ESP8266. Bulletin of Electrical Engineering and Informatics, 11(6), 3403–3413. <https://doi.org/10.11591/eei.v11i6.4360>